

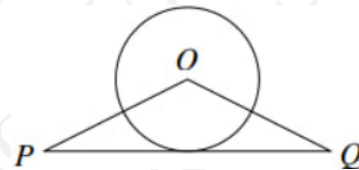
NUET Mock №7

Kiseki no Mock

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Mathematics

1. The diagram shows a circle with centre O and a triangle OPQ . Side PQ is a tangent to the circle. The area of the circle is equal to the area of the triangle.



What is the ratio of the length of PQ to the circumference of the circle

- A. 1 : 1
- B. 2 : 3
- C. 2 : π
- D. 3 : 2
- E. π : 2

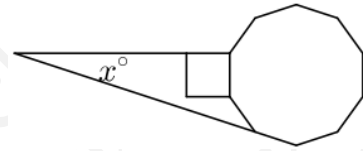
Ans: A

2. The positive integer k satisfies the equation $\sqrt{2} + \sqrt{8} + \sqrt{18} = \sqrt{k}$. What is the value of k ?

- A. 28
- B. 36
- C. 72
- D. 128
- E. 288

Ans: C

3. The diagram shows a square and a regular decagon that share an edge. One side of the square is extended to meet an extended edge of the decagon.



What is the value of x ?

- A. 15
- B. 18
- C. 21
- D. 24
- E. 27

Ans: B

4. A square has vertices $(0, 0)$, $(1, 0)$, $(1, 1)$ and $(0, 1)$. Graphs of the following equations are drawn on the same set of axes as the square.

$$x^2 + y^2 = 1, \quad y = x + 1, \quad y = -x^2 + 1, \quad y = x, \quad y = \frac{1}{x}$$

How many of the graphs pass through exactly two of the vertices of the square?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

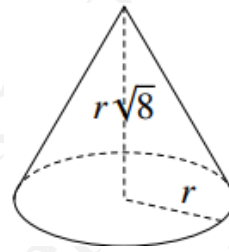
Ans: C

5. When Louise had her first car, 50 litres of petrol cost £40. When she filled up the other day, she noticed that 40 litres of petrol cost £50. By approximately what percentage has the cost of petrol increased over this time?

A. 50%
B. 56%
C. 67%
D. 75%
E. 80%

Ans: B

6. A solid right circular cone whose base has radius r metres has total surface area (including the circular base) of A metres squared and a volume of V metres cubed. The perpendicular height of the cone is $r\sqrt{8}$ metres.



Given that $\frac{V}{A} = \sqrt{5000}$, what is the value of r ?

A. 200
B. 225
C. 250
D. 275
E. 300

Ans: E

7. The parabola $y = x^2 - 10x + 24$ has two distinct x -intercepts and a y -intercept. The circle which goes through these three intercepts meets the parabola at a fourth point, P .

What is the distance of P from the origin?

- A. 26
- B. $26\sqrt{2}$
- C. 28
- D. $30\sqrt{3}$
- E. 35

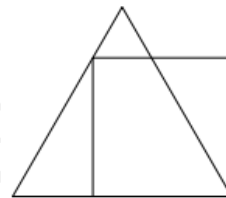
Ans: A

8. The equations $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ both have real roots. It is known that the sum of the squares of the roots of the first equation is equal to the sum of the squares of the roots of the second one, and that $a \neq b$. Find the value $500 + 10(a + b)$.

- A. 190
- B. 240
- C. 360
- D. 480
- E. 500

Ans: D

9. The perimeter of the square in the figure is 40. The perimeter of the larger equilateral triangle in the figure is $a + b\sqrt{p}$, where p is a prime number. What is the value of $7a + 5b + 3p$?



- A. 269
- B. 301
- C. 357
- D. 441
- E. 463

Ans: A

10. A dealer bought two cars. He sold the first one for 40% more than he paid for it and the second one for 60% more than he paid for it. The total sum he received for the two cars was 54% more than the total sum he paid for them. When written in its lowest terms, the ratio of the prices the dealer paid for the first and the second car was $a : b$. What is the value of $11a + 13b$?

- A. 86
- B. 102
- C. 124
- D. 172
- E. 198

Ans: C

11. The point A is due west of the point B.
The bearing of the point C from B is 150° .
The distance between point A and point B is the same as the distance between point B and point C.
A is 16 km away from C.
Find the distance of AB.

- A. 16 km
- B. $\frac{16}{\sqrt{3}}$ km
- C. $16\sqrt{3}$ km
- D. 8 km
- E. $\frac{16}{\sqrt{2}}$
- F. $\frac{16\sqrt{3}}{2}$

Ans: B

12. A woman has 31 coins, all of which are dimes (\$0.1) and quarters (\$0.25). If the total value of the coins is \$5.95, by how much more the value of all the quarters is compared to the value of all the dimes?

A. \$2.75
B. \$3.00
C. \$3.25
D. \$3.55
E. \$3.75
F. \$4.00

Ans: D

13. The expression $\sqrt{\frac{16^x+8^x}{4^x+2^x}}$ simplifies to which of the following?

A. 2^x
B. 2
C. $(\sqrt{2})^x$
D. $\sqrt{2^x}$
E. 2^{2x}

Ans: A

14. Which one of the following is an equation of the line that passes through (4, 3) and perpendicular to the line $y = 2x + 4$?

A. $2x - y = 5$
B. $2y + x = 11$
C. $2y - x = 2$
D. $2x + y = 11$
E. $2y + x = 10$

Ans: E

15. A car starts at point X. It travels 3.0 km due east, then 4.0 km due south, then 6.0 km due west and finally 8.0 km due north.

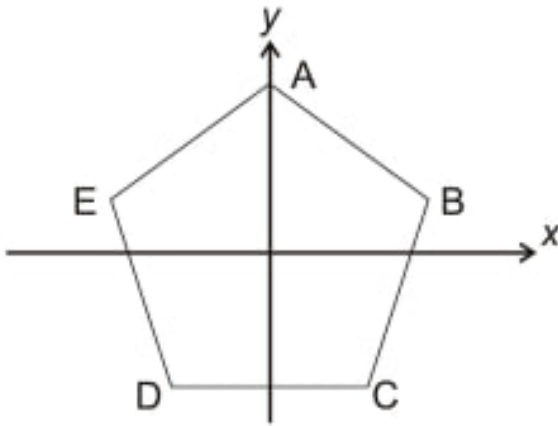
How far away is the car from point X when it has reached the end of this journey?

(Assume that all distances moved are on a flat horizontal surface, and that point X is on the equator. You may ignore any curvature of the Earth.)

- A. 5.0 km
- B. 21.0 km
- C. 10.0 km
- D. 7.0 km
- E. 2.0 km

Ans: A

16. ABCDE is a regular pentagon. The transformation R is a rotation about the origin and maps A to B, B to C, C to D, etc. The transformation S is a reflection in the y-axis.



Which of the following sequences of transformations (performed in the order that they are listed) would NOT leave vertex D in the same position?

- A. S R R S R S
- B. S R
- C. R R S R S R
- D. R S R S
- E. R S R R

Ans: C

17. A group of drivers, consisting of 200 women and 300 men, was asked if they passed their driving test at the first attempt.

Altogether 167 of the group said they passed at the first attempt.

Of the women, 143 said they did not pass at the first attempt.

How many of the men said they passed at the first attempt?

- A. 10
- B. 24
- C. 33
- D. 57
- E. 110
- F. 133
- G. 157

Ans: E

18. How many different integers, n , are there such that the difference between $2\sqrt{n}$ and 7 is less than 1?

- A. 0
- B. 2
- C. 4
- D. 6
- E. 8

Ans: D

19. Four individual spheres have radii:

$\frac{r}{2}$, r , $2r$ and $3r$.

What is the sum of their surface areas?

- A. $58\pi r^2$
- B. $169\pi r^2$
- C. $57\pi r^2$
- D. $25\pi r^2$
- E. $26\pi r^2$

Ans: C

20. The quantities x and y are positive.

x is inversely proportional to the square root of y .

When $x = 8$, $y = 9$.

What is the value of y when $x = 6$?

- A. $\frac{3}{2}$
- B. 2
- C. $\frac{81}{16}$
- D. $\frac{27}{14}$
- E. 12
- F. 16

Ans: F

21. Solve the inequality:

$$\frac{1}{3}(x+2) - \frac{3}{4}(x-1) > 2-x$$

- A. $x < -2$
- B. $x \leq 0$
- C. $x > 1$
- D. $x \geq 2$
- E. $x = 3$

Ans: C

22. P is a regular 12-sided polygon.

The length of a straight line joining a vertex to the center of P is 6 cm.

Q is a regular hexagon with side length x cm.

The areas of P and Q are equal.

What is the value of x^2 ?

- A. 12
- B. 24
- C. 36
- D. 72
- E. $12\sqrt{3}$
- F. $24\sqrt{3}$
- G. $36\sqrt{3}$
- H. $72\sqrt{3}$

Ans: F

23. In a rectangle, the long side is 3 times greater than the short side. If we cut 1 cm from each side, the perimeter will be 36 cm. Find the area of the rectangle at first.

A. 45
B. 60
C. 75
D. 80
E. 84

Ans: C

24. On Aigerim's 18th birthday in 2012, she was 159 cm in height, having grown 6% since her birthday in 2011. How tall was she in 2011?

A. 148 cm
B. 145 cm
C. 150 cm
D. 135 cm
E. 152 cm

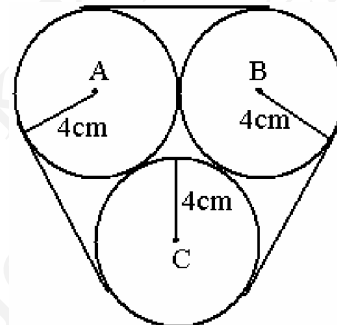
Ans: C

25. In the given figure, three circles centered at A, B, and C have equal radii of 4 cm.

The circles are tangent to each other, and a metal wire rounds the circles completely as shown in the figure.

Find the length of the metal wire.

Given that $\frac{V}{A} = \sqrt{5000}$, what is the value of r ?



A. $4(\pi + 2)$
B. $4(\pi + 3)$
C. $8(\pi + 3)$
D. $8(\pi + 2)$
E. $16(\pi + 3)$

Ans: C

26. David drove at an average speed of 50 km/h for a distance of 45 km in his car. David's journey took 54 minutes.

On a different journey, Marian drove at an average speed of 70 km/h. The time she took on her journey was 80% more than the time taken by David on his journey.

What is the distance, in km, that Marian drove?

- A. 56 km
- B. 113 km
- C. 85 km
- D. 216 km
- E. 52 km

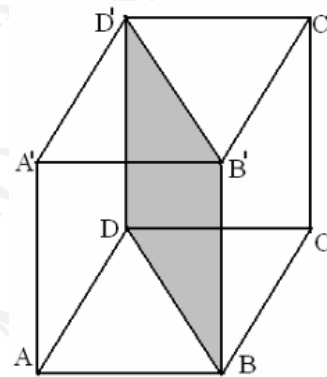
Ans: B

27. Convert 16 km/h to m/s.

- A. 5 m/s
- B. $\frac{20}{3}$ m/s
- C. $\frac{40}{9}$ m/s
- D. 10 m/s
- E. 8 m/s

Ans: C

28. The volume of the given cube is 64 cm^3 . Find the area of the given rectangle $DBB'D'$.



- A. $8\sqrt{2}$
- B. $10\sqrt{2}$
- C. $12\sqrt{2}$
- D. $16\sqrt{2}$
- E. $24\sqrt{2}$

Ans: D

29. A metal statue is melted down and turned into 125 smaller statues. The new statues are all identical to each other and are scaled down versions of the original.

The area of the base of one of the new statues is 100 cm^2 .

What is the area of the base of the original statue?

- A. 500
- B. 2500
- C. 5000
- D. 2000
- E. 3000

Ans: B

30. What is the surface area of a regular square pyramid with a slant height of 12 cm and a lateral edge of 13 cm?

- A. 240
- B. 450
- C. 600
- D. 100
- E. 340

Ans: E

Critical Thinking

1. The town of Coppersville uses some special coins. Edwards has five 5 p coins, four 4 p coins, three 3 p coins, two 2 p coins and one 1 p coin. The smallest amount he cannot pay with his coins alone is W p.

What is the value of W ?

- A. 50
- B. 56
- C. 47
- D. 33
- E. 25

Ans: B

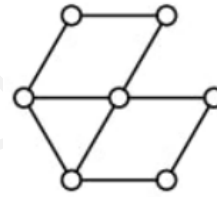
2. One of the best things about online courses is that they are accessible to people all over the world. However, because there are so many courses available, finding one that is genuinely high-quality can be just as time-consuming as enrolling in a traditional class. Additionally, there's no guarantee that the course material is accurate or up-to-date since anyone can create an online course without proper accreditation or review. Therefore, attending accredited institutions in person is a far better option for those who are serious about their education.

Which one of the following is an assumption underlying the above argument?

- A. Most online courses contain outdated or incorrect information.
- B. Traditional classes are more expensive than online courses.
- C. Accredited institutions are more likely to provide reliable and accurate course material.
- D. Online courses are popular mainly because they are convenient.
- E. Many online course creators intentionally provide misleading information.

Ans: C

3. The circles in the diagram are to be coloured so that any two circles connected by a line segment have different colours.

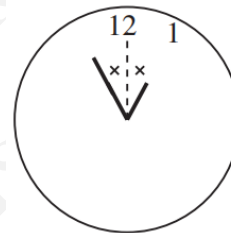


What is the smallest number of colours required?

- A. 6
- B. 5
- C. 4
- D. 3
- E. 2

Ans: D

4. Priti is learning a new language called Tedio. During her one hour lesson, which started at midday, she looks at the clock and notices that the hour hand and the minute hand make exactly the same angle with the vertical, as shown in the diagram. How many whole seconds remain until the end of the lesson?



- A. 235
- B. 276
- C. 294
- D. 311
- E. 321

Ans: B

5. In the 1980s, there was a public transport strike in a major city. It was predicted that because public transport was unavailable, there would be a significant increase in traffic accidents due to more people using private cars. However, the data showed that the number of traffic accidents actually decreased during the strike. From this, we can conclude that using public transport, rather than reducing road accidents, may actually contribute to them.

Which one of the following states a flaw in the above argument?

- A. It ignores other possible reasons for the decrease in traffic accidents.
- B. It implies that public transport should be banned to reduce accidents.
- C. It presents data that contradicts its own conclusion.
- D. It assumes a direct link between public transport usage and accident rates.
- E. It assumes that predictions about traffic accidents were incorrect.

Ans: A

6. Although it has long been known that dolphins communicate through clicks and whistles, advances in marine biology have revealed that dolphins use a complex system of sounds to coordinate hunting and social interactions. Using underwater microphones, researchers have discovered that dolphins have unique signature whistles, functioning like names, which help them identify and call each other over long distances. However, these sophisticated communication abilities can be exploited by humans. For instance, some fishing vessels use devices that mimic dolphin calls to attract them, making dolphins vulnerable to capture or harm. This has raised concerns among conservationists about the unintended consequences of human interference with dolphin communication.

Which of the following can be drawn as a conclusion from the passage above?

- A. The use of underwater microphones has helped discover new dolphin species.
- B. Dolphins are the only marine animals that use sound for identification.
- C. Dolphins have developed complex communication solely to avoid predators.
- D. Mimicking dolphin sounds is an effective way to protect them from human threats.
- E. Dolphins' communication skills will eventually lead to their vulnerability to human activities.

Ans: E

7. The following table shows the exchange rates for five world currencies against the UK pound. An investor will change some pounds into one of these currencies in May and exchange the full amount back into pounds in October.

Currency	May	June	July	August	September	October
US dollar	2.1	2	1.8	1.7	1.5	1.6
SINGAPORE dollar	1.4	1.2	1.3	1.25	1.4	1.35
BRUNEI dollar	2.4	2.5	2.6	2.6	2.7	2.8
AUSTRALIAN dollar	2	2.1	2	1.95	1.9	1.7
ARGENTINE peso	6.3	6.2	5.9	4.5	5	5.2

For example, 1 UK pound buys 2 AUSTRALIAN dollars in May.

Which currency will make the investor the most money in UK pounds over the six months?

- A. ARGENTINE peso
- B. AUSTRALIAN dollar
- C. BRUNEI dollar
- D. SINGAPORE dollar
- E. US dollar

Ans: E

8. Stephanie enjoys swimming. She goes for a swim on a particular date if, and only if, the day, month (where January is replaced by '01' through to December by '12') and year are all of the same parity (that is they are all odd, or all are even). On how many days will she go for a swim in the two-year period between January 1st of one year and December 31st of the following year inclusive?

- A. 142
- B. 155
- C. 183
- D. 203
- E. 216

Ans: C

9. Either people are becoming healthier due to improved lifestyles, or medical treatments have become more effective at curing diseases. Since there is little evidence that lifestyle habits such as diet and exercise have significantly improved in recent years, we must conclude that medical treatments are more effective now than they were in the past.

Which one of the following most closely parallels the reasoning used in the above argument?

- A. Farmers must decide whether to plant wheat or corn based on seasonal weather conditions. Since weather forecasts predict more rain this year, they should choose corn.
- B. Every year, students choose between attending public or private universities. Since private universities often have better facilities, many students prefer them.
- C. Researchers studying bird migration found that birds either rely on instinct or environmental cues to navigate. Since experiments show that removing environmental cues disrupts migration, it must be the key factor.
- D. Wood-burning stoves are considered more eco-friendly than traditional heating methods because they produce less carbon emissions. Therefore, more people should adopt them.
- E. When scheduling surgeries, doctors prioritize critical cases over routine procedures. If your case is not critical, you may have to wait longer.

Ans: C

10. In a situation where every possible choice leads to some form of harm, it is still possible to choose the option that minimizes harm while maximizing overall benefit. Therefore, when faced with two harmful decisions, one should select the option that causes the least damage while providing the greatest benefit to the largest number of people. This may be a difficult conclusion to reach, but it remains the most ethical course of action.

Which one of the following best illustrates the principle contained

- A. The quickest and least costly decision is often the most ethical one.
- B. The decision that causes the least harm to the fewest people is the right choice.
- C. An action is not unethical if it achieves the intended outcome efficiently.
- D. When all options result in harm, it is important to choose the option with the least harmful consequences.
- E. An action cannot be considered unethical if it is done with good intentions.

Ans: D

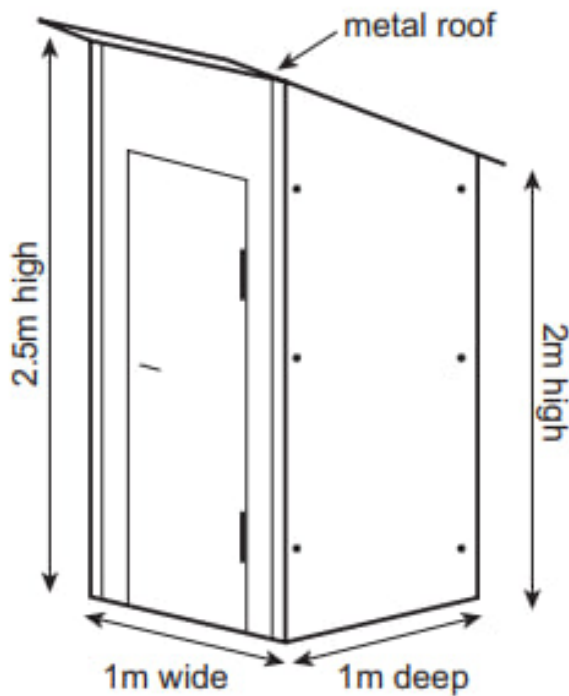
11. Luxury cruises on very large ships have become an increasingly fashionable leisure activity, especially among retired people. However, people should not put themselves in danger unnecessarily. Some of the most popular cruises go through parts of the oceans where modern pirates are active. Passengers face a serious risk of being captured and held for ransom or even being killed when the ship is invaded. In addition, it is wrong to benefit from the exploitation of workers. The workers in the engine rooms, laundries, kitchens and restaurants of large cruise ships work long hours for poor pay under appalling conditions and in a culture of fear, facing instant dismissal if they complain about bullying or sexual harassment.

Which one of the following can be drawn as a conclusion from the above passage?

- A. Most of the people who can afford the time and cost of luxury cruises are those who have retired from work.
- B. The governments of the world should work together to eradicate the danger of piracy.
- C. There are personal and moral reasons why people should not go on luxury cruises.
- D. Owners of luxury cruise ships operate in a fiercely competitive business environment
- E. Workers on large cruise ships should band together to campaign for better pay and working conditions.

Ans: C

12. The drawing below shows an allotment shed.



The four outside wood surfaces of the shed, including the door, need to be painted with three coats of paint. It takes 4 minutes to paint 1 square metre.

I have two identical sheds to paint and will take just one 20-minute tea break.

How long will it take to paint the two sheds?

- A. 36 minutes
- B. 1 hour and 48 minutes
- C. 2 hours and 8 minutes
- D. 3 hours and 36 minutes
- E. 3 hours and 56 minutes

Ans: E

13. Sarah drives to work each morning. On the way she passes through a busy junction with a set of traffic lights. One morning she stops behind a line of traffic just as the lights go red. She can see 14 cars between her and the red light. The red (stop) light will remain on for 42 seconds. The green (go) light will come on for just 10 seconds, and this will be enough time for 7 cars to get through the junction before the red light comes on again and the line of cars is brought to a stop.

How long will Sarah have to wait until she can go through the junction?

- A. 52 seconds
- B. 1 minute 34 seconds
- C. 1 minute 44 seconds
- D. 2 minutes 26 seconds
- E. 2 minutes 36 seconds

Ans: D

14. Government legislation has proved very effective in reducing emissions from car exhausts. But exhausts are not the only source of air pollution linked to traffic. A recent study showed that particles from brakes, tyres and road surfaces actually made up about two thirds of all dangerous particulate matter from road transport. It is now time to legislate to impose lower speed limits in areas where there are high volumes of traffic.

Which one of the following is an assumption underlying the above argument?

- A. Lower speed limits will lead to a reduction in car exhaust emissions.
- B. There are no other sources of air pollution besides those caused by traffic.
- C. Air pollution is only a problem in areas where there is a high volume of traffic.
- D. If legislation has been effective elsewhere then it is likely to be effective in this case.
- E. Pollution from brakes, tyres and road surfaces is reduced when lower speed limits are imposed.

Ans: E

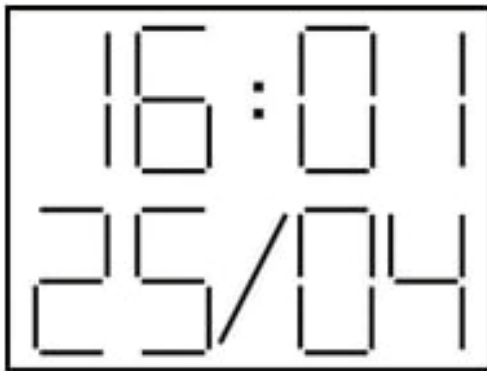
15. Risk of death contradicts the very concept of sport, since sport is generally considered to belong to the less serious side of life. Sport is what people do to counter the stress and pressure of work, not to increase them. The growth in the popularity of so-called extreme sports, where there is a real risk of death, is therefore puzzling; particularly since the modifications in equipment which are designed to make the sports safer do not lead to fewer accidents but to ever higher performance levels and greater exposure to danger.

Which one of the following, if true, most weakens the above argument?

- A. Improved performance in extreme sports does not lessen the risk of death.
- B. Most people want to escape from the modern obsession with safety at work.
- C. By claiming to enhance safety, sports-equipment designers merely encourage risk-taking.
- D. The popularity of extreme sports means that sport is sometimes a serious matter.

Ans: B

16. I have a digital clock that displays the date as well as the time. Just after 4 p.m. on 25th April I happened to glance at the clock and saw:



It struck me that the four numbers on the clock were all different square numbers. When I thought about it, I realised that this would occur another three times within the next hour, then not again until September.

How many times will my clock display four different non-zero square numbers during September?

- A. 32

- B. 36
- C. 40
- D. 44
- E. 48

Ans: B

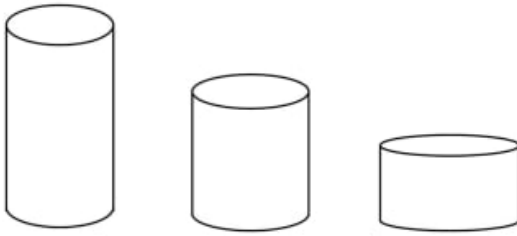
17. The battery life of laptop computers means that they cannot be used for the duration of a long haul flight without needing to be recharged from an onboard power point. Not all aeroplanes have this facility. Therefore, laptop computers should not be allowed as carry-on items for long haul flights on planes that do not have onboard power points, regardless of whether they are intended for personal or business use, as they will be of limited use to their owners and take up a disproportionate amount of space in the cabin.

Which one of the following is the best statement of the flaw in the above argument?

- A. Some companies expect their employees to work on long haul flights.
- B. There is no chance of a laptop becoming damaged if it is stored in a suitcase in the hold.
- C. A laptop may run out of battery on a short haul flight, too.
- D. The passenger cannot choose whether or not they book a flight with an onboard power point.
- E. Passengers could make some use of their laptop even if the battery does run out.

Ans: E

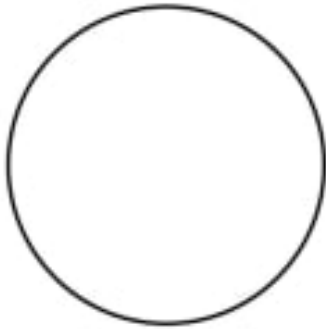
18. Below are pictures of three cylinders.



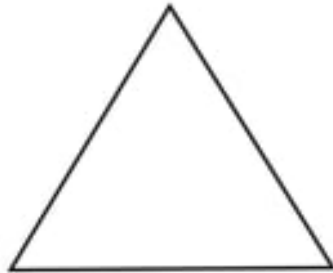
If a cylinder is cut into two equal halves with one straight cut, which one of the following could NOT be a view of one of the halves from any direction?



A.



B.



C.



D.



E.

Ans: C

19. Myopia (short-sightedness) has long been thought of as a genetic condition. However, recent research suggests that environmental factors are as important. In particular, the idea that unusually large amounts of reading contribute to the development of myopia is supported by a recent academic study of 4568 people aged 35–74 years. This showed that 50% of those who went to university were myopic compared to 34% of non-graduates. So, if you want to reduce your chances of developing myopia you should not go to university.

Which one of the following is an underlying assumption of the above argument?

- A. People who go to university have to do a lot of reading.
- B. Myopia is a serious eye condition which one should avoid.
- C. It is not worth going to university.
- D. There are no other ways to reduce the risk of developing myopia.
- E. Lenses to correct myopia are not fully effective

Ans: A

20. Jane and Tom make wedding cakes. Jane was paid £240 by a customer. Tom bought the ingredients for the cake, which came to £40. They will split the costs of all ingredients equally, but Jane will keep 60% of the customer's fee, compared to Tom's 40%. How much does Jane owe Tom?

A. £96
B. £112
C. £116
D. £120
E. £140

Ans: C

21. In a certain family, the father gives out pocket money to his children in the following way. Every month he chooses a number between 1 and 100, and then shares that amount of money equally between the seven children, giving them whole-number sums of money, and giving anything left over to charity.

For instance, if he chose the number 30, he would give \$4 to each child ($30 \div 7 = 4$ with a remainder of 2) and \$2 to charity.

If the father realizes after 100 months that every number between 1 and 100 (inclusive) has been used once, how much has he given to charity?

A. \$276
B. \$294
C. \$295
D. \$297
E. \$300

Ans: B

22. Alice and Ben both carry pedometers when walking to monitor how far they go. One day they go the same route, which Alice completes in 53 minutes, while Ben only takes 45 minutes. Alice's pedometer registers the route length as 5.27 km, but Ben's pedometer suggests that it is 5.69 km. Neither knows whether their pedometer is accurate, so Alice takes hers to a reliable expert who tells her that her pedometer has been under-estimating distances travelled by 3.26%. Ben knows that his pedometer may consistently under or overestimate distances but does not get it checked.

Ben wishes to calculate as accurately as possible his average speed on the route he took with Alice, and his average speed on future walks he takes with his own pedometer. Which one of the pieces of information given above is **unnecessary** to complete such calculations?

- A. Alice completed the route in 53 minutes.
- B. Alice's pedometer registered the route as 5.27 km long.
- C. Alice's pedometer under-estimates distance by 3.26%.
- D. Ben completed the route in 45 minutes.
- E. Ben's pedometer registered the route as 5.69 km long.

Ans: A

23. A recent study has found that those businesses which engaged more with social media had more financial success than businesses which did not. The businesses which were the most engaged saw their revenue grow by 18 per cent over the last year, while the least engaged saw losses of 8 per cent. Since social media is the future of communication, it can be seen that businesses should interact more with social media in order to maximise their profits.

Which one of the following is the best statement of the flaw in the above argument?

- A. It ignores the fact that not all businesses use social media.
- B. It assumes that businesses that do not use social media will collapse.
- C. It assumes that participation in social media causes an increase in profits.
- D. It ignores the fact that businesses can use other forms of communication.
- E. It assumes that a growth in revenue of 18 per cent is possible for all businesses.

Ans: C

24. It is time for prostitution to be legalised. It has been illegal for millennia, and yet it is still present in modern society. Clearly, its being illegal does not provide enough deterrent to prevent people from engaging in it. We are never going to eradicate prostitution, so we should give up trying, and legalise it. Money that is spent in prosecuting prostitutes and their clients could be better used elsewhere.

Which one of the following illustrates the principle used in the above argument?

- A. Prostitutes' clients should be prosecuted, but not prostitutes themselves.
- B. Some people will commit murders, whether or not murder is illegal, so murder should not be illegal.
- C. People who are involved with dealing in illegal substances largely as a result of a background of poverty should not be prosecuted.
- D. Modern crimes, such as internet hacking, should be the focus of police forces.
- E. We should not waste money trying to rehabilitate young offenders.

Ans: B

25. In order for it to be possible to rain, clouds must be present. The weather forecast is predicting clear skies, so there is no chance of rain.

Which one of the following most closely parallels the reasoning used in the above argument?

- A. The temperature in the fish tank must be at least 25°C for the fish to survive. The fish are all alive, therefore the temperature must be at least 25°C.
- B. Solar eclipses only occur at New Moon, when the Moon is between the Earth and the Sun. The New Moon is not until next week, so there will be no Solar eclipse today.
- C. A scab will only form when platelets in the blood have clotted. If you do not have any scabs, you do not have platelets in your blood.
- D. When outside temperatures fall below 0°C household pipes are at risk of bursting. So at a temperature of -5°C all your household pipes will burst.
- E. Deciduous trees lose their leaves in the autumn. Autumn will be late this year, so trees will not lose their leaves until winter.

Ans: B

26. I live in a block of flats which is served by a lift which always goes from bottom to top and back, picking up anyone who wishes to go in the direction it is travelling. If I have pressed the 'down' button and the lift is going up, it does not stop for me, but goes to the top and stops for me on the way down.

I notice that, if I have pressed the button on my floor to go up, more often than not the lift goes past my floor going down before it comes back up to collect me.

Which one of the following would explain this?

- A. The lift travels down faster than it travels up.
- B. It waits longer at the bottom floor than at intermediate floors.
- C. People mainly travel between their own floor and the bottom.
- D. I live near the top of the block of flats.
- E. I live near the bottom of the block of flats.

Ans: E

27. Many people criticise what they call 'the younger generation' for what their seniors regard as immoral behaviour. These critics need to realise that because the world has changed, moral standards have changed with it. In fact, technology is advancing so rapidly that it is only the young who understand the world in which they live. It is they, not their parents or grandparents, who should be setting moral standards. Anyone over the age of 25 cannot help but be blinded by what they learned in the past. Their education has prevented them from perceiving and responding to the demands of the modern day, and so their moral judgments have no validity.

Which one of the following best expresses the main conclusion of the above argument?

- A. Moral judgments made by people over the age of 25 are not valid.
- B. Moral standards have changed.
- C. Moral standards should be set by people under the age of 25.
- D. The younger generation is guilty of what older people regard as immoral behaviour.
- E. Young people are the only ones who understand the contemporary world.

28. Anna buys potatoes in the local market. She has noticed that the price on Saturday morning is 5p per kilo more than the normal weekday price. However, an hour before the stall closes on a Saturday afternoon, the price drops to 5p per kilo below the weekday price. She spends £3.00 a week on potatoes. This will buy her 3 kilos less in weight on a Saturday morning than on a weekday, but 5 kilos more in weight at the low Saturday afternoon price than on a weekday.

What is the weekday price of potatoes per kilo?

- A. 12p
- B. 15p
- C. 20p
- D. 25p
- E. 30p

Ans: C

29. The payment systems which use pre-paid magnetic swipe cards are not all that wonderful. We had one put into the office last week to operate the tea and coffee dispensing machine. I was somewhat surprised to find, after having five coffees, that I had spent 121p, and my colleague, after four teas, had spent 82p. We were assured that there had been no changes in the prices; the problem appears to be that if you ask the machine how much you have left, it deducts a 'service' charge for this 'service'. We both enquired once. Charges for tea, coffee, and service are whole numbers.

What is the minimum that the service charge could be?

- A. 1p
- B. 2p
- C. 4p
- D. 6p
- E. 14p

Ans: D

30. There is no point to re-enactments of historical battles. Despite the meticulous attention to getting the uniforms right and the cannon smoke, the simulation of being there fails on every level: there is no real danger of death and you know what the outcome will be. It is merely an exercise in pageantry. In history-as-lived, the participants at Gettysburg did not know that the Confederates were not going to win, while in a scripted re-enactment everyone knows their role and destiny from the outset.

Which one of the following best expresses the main conclusion of the above argument?

- A. Re-enactments of historic battles are just colourful pageants.
- B. The simulation of being there at a re-enacted historic event fails on all levels.
- C. Re-enactments of historic battles are pointless.
- D. No insights into historic battles are gained through re-enactments.
- E. Pageantry is of no value in itself.

Ans: C